

AI for Economic Research: Dynamic Models, Language, and Agents

Lecture 1.3a: Model Extensions and Theoretical Challenges

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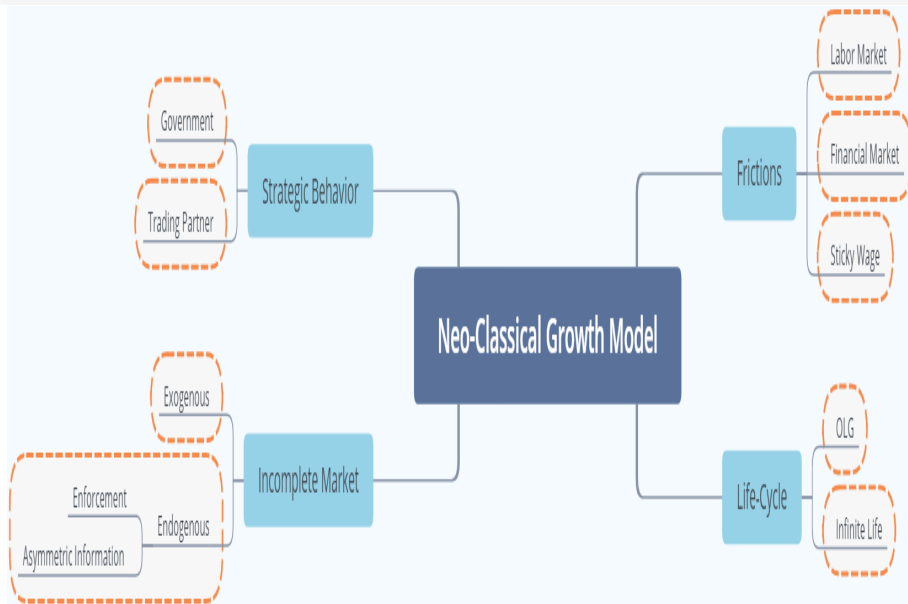
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Topic 1.3a Agenda

1. **Heterogeneous Agent Models** — Aiyagari, Krusell–Smith
2. **Theoretical Issues** — existence, uniqueness, continuity
3. **Curse of Dimensionality** — the computational wall
4. **Numerical Methods** — interpolation, optimization, simulation
5. **Calibration, Estimation, Accuracy** — mapping models to data
6. **Perturbation Methods** — and supplementary video resource

Model Extensions and Challenges

Extensions Based on the Optimal Growth Model



Heterogeneous Agents: Breaking the Simplicity

- **Realistic Extension:** Agents differ in wealth, income, productivity
 - Individual state: (a_i, z_i) — assets and idiosyncratic shock
 - Now $k_i \neq K$; individual choices don't reveal aggregates
- **Prices Depend on the Distribution:**

$$r = r(\mu), \quad w = w(\mu)$$

where μ is the *cross-sectional distribution* of (a, z)

- **The Distribution Is a State Variable:**
 - To forecast prices, agents must forecast how μ evolves
 - But μ is *infinite-dimensional*—a measure over (a, z)
 - Evolution: $\mu' = \Phi(\mu; g)$ depends on everyone's policy
- **The Environment Is No Longer Obvious:**
 - Agents face a *perceived law of motion* they must approximate
 - First Welfare Theorem fails $\Rightarrow$ must solve for prices explicitly

Bewley-Aiyagari-Huggett Model

- **Setup:**

- Continuum of infinitely lived households (measure 1)
- Uninsurable idiosyncratic income shocks ϵ
- *Ex-ante* identical, *ex-post* heterogeneous

- **Bellman Equation:**

$$v(a, \epsilon; \lambda) = \max_{c, a'} \left\{ u(c) + \beta \sum_{\epsilon'} v(a', \epsilon'; \lambda') \pi(\epsilon, \epsilon') \right\}$$

subject to:

$$c + a' = (1 + r(\lambda))a + w(\lambda)\epsilon$$

$$a' \geq -\underline{a}$$

- **Stationary Equilibrium:** Find λ^* such that $\lambda^* = \Phi(\lambda^*; g)$

- Distribution is invariant under optimal policies
- Incomplete markets endogenously generate wealth inequality

Krusell-Smith Model

- **Extension:** Add *aggregate uncertainty* to Aiyagari
- **Aggregate shock:** $z_t \in \{z_g, z_b\}$ (good/bad states)
 - Affects aggregate productivity
 - Unemployment higher in bad states: $u_b > u_g$

- **Household Problem:**

$$v(k, \epsilon; \Gamma, z) = \max_{c, k'} \{U(c) + \beta \mathbb{E}[v(k', \epsilon'; \Gamma', z') \mid z, \epsilon]\}$$

subject to:

$$\begin{aligned} c + k' &= r(\bar{k}, \bar{l})k + w(\bar{k}, \bar{l})\epsilon + (1 - \delta)k \\ \Gamma' &= H(\Gamma, z, z'), \quad k' \geq 0 \end{aligned}$$

- **The Challenge:** Agents must forecast evolution of Γ —infinite-dimensional!

Krusell-Smith: The Computational Innovation

- **Key Insight (Krusell-Smith, 1998):**

- Agents don't need to track the entire distribution Γ
- Approximate with low-dimensional moments: $\bar{K} = \int k d\Gamma$

- **Bounded Rationality:** Agents use simple forecasting rules

$$\log \bar{K}' = a_0(z) + a_1(z) \log \bar{K}$$

- **Algorithm:**

1. Guess coefficients (a_0, a_1)
2. Solve individual problem given forecasting rule
3. Simulate economy, regress $\log \bar{K}'$ on $\log \bar{K}$
4. Update coefficients; iterate until convergence

- **Remarkable Finding:** $R^2 > 0.999$ —mean capital is nearly sufficient!

Theoretical Issues in Heterogeneous Agent Models

- **Existence of Recursive Equilibrium:**
 - When markets are complete/efficient: easier to prove
 - Incomplete markets: may require forward-looking variables
- **Uniqueness:** Generally *not* guaranteed
 - Multiple steady states possible
 - Policy functions may have discontinuities
- **Continuity:** Value and policy functions may not be continuous in aggregates
- **Computation:**
 - Curse of dimensionality (distribution as state)
 - Ensuring accuracy of approximations
 - Verifying equilibrium conditions
- **These issues become more severe in optimal policy (Ramsey) problems.**

Example: Non-Existence of Continuous Equilibrium

Santos (2002): Representative household, $\sum \beta^t \log(c_t)$, subject to:

$$c_t + k_{t+1} \leq \pi_t + (1 - \tau_t)r_t k_t + T_t$$

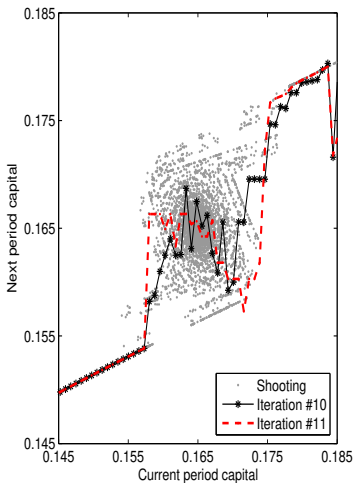
with piecewise linear tax on capital:

$$\tau(K) = \begin{cases} 0.10 & \text{if } K \leq 0.160 \\ 0.05 - 10(K - 0.165) & \text{if } 0.160 < K < 0.170 \\ 0 & \text{if } K \geq 0.170 \end{cases}$$

Result: A continuous Markov equilibrium *fails to exist*.

- Middle steady state: unstable (complex eigenvalues)
- Other steady states: saddle-path stable
- Equilibrium exhibits cycles or discontinuities

Theoretical Issues: Equilibrium Cycles



Ramsey Problem: Additional State Variables

- **Consumer FOCs:**

- Consumption: $\lambda_t = u_{c_t}$
- Labor: $u_n = (1 - \tau_t)u_c f_n$
- Euler: $\lambda_t = \lambda_{t+1}\beta[(1 - \theta_{t+1})f_k + 1 - \delta]$

- **The Problem:** Euler equation can be rewritten backward:

$$\lambda_{t-1} = \lambda_t \beta [(1 - \theta_t) f_k(k_t, n_t) + 1 - \delta]$$

- **Implication:** Must track λ_{t-1} as a state variable

- Minimal state space (k_t) is *insufficient*
- Need (k_t, λ_{t-1}) with unknown domain for λ

- **Consequence:** State space expands; recursive formulation becomes more complex.

The Curse of Dimensionality

- **Bellman (1957):** Named the “curse of dimensionality”—exponential growth of computational cost with state dimension.
- **Grid-Based Methods:**
 - n grid points per dimension, d dimensions $\Rightarrow n^d$ total points
 - Example: 100 points $\times$ 10 dimensions = 10^{20} points
 - Infeasible for storage, let alone computation
- **Heterogeneous Agent Models:**
 - Distribution μ is infinite-dimensional
 - Even discretized: thousands of (a, z) pairs $\times$ aggregate states
 - Standard VFI becomes computationally prohibitive
- **This is exactly the challenge AI/RL research confronts**—in even more extreme settings.

Numerical Methods: Interpolation and Approximation

- **Purpose:** Represent value and policy functions on a computer.
- **Interpolation Methods:**
 - *Linear interpolation:* Simple, fast, not smooth
 - *Cubic splines:* Smooth, accurate for smooth functions
 - *Chebyshev polynomials:* Near-optimal approximation on $[-1, 1]$; avoid Runge's phenomenon
- **Discretization of Stochastic Processes:**
 - *Tauchen (1986):* Discretize AR(1) $z_t = \rho z_{t-1} + \sigma \varepsilon_t$ via finite Markov chain
 - *Rouwenhorst (1995):* Better for highly persistent processes ($\rho \approx 1$)
- **Projection Methods:**
 - Approximate $v(k) \approx \sum_j c_j \phi_j(k)$ using basis functions
 - Choose coefficients to minimize residuals in Bellman equation

Numerical Methods: Optimization

- **Maximization in the Bellman Equation:**

$$\max_{k' \in \Gamma(k)} \{u(f(k) - k') + \beta v(k')\}$$

- **Classical Methods:**

- *Grid search*: Evaluate objective at all grid points; pick max — reliable but slow
- *Golden section search*: Exploit unimodality; $O(\log n)$ evaluations
- *Newton-Raphson*: Use derivatives for fast convergence; $O(\text{quadratic})$

- **Modern Methods:**

- *Automatic differentiation (PyTorch)*: Exact gradients from code
- *Stochastic gradient descent (SGD/Adam)*: Scale to high dimensions
- *Gradient-based optimization*: Essential for neural network training

- **Rule of Thumb**: Use derivatives when available; they dramatically speed up convergence.

Numerical Methods: Simulation

- **Monte Carlo Simulation:** Generate artificial time series from a calibrated model.
- **Algorithm:**
 1. Draw shocks $\{z_t\}_{t=1}^T$ from the Markov chain
 2. Given policy g , compute $k_{t+1} = g(k_t, z_t)$ and $c_t = f(k_t, z_t) - k_{t+1}$
 3. Compute moments: mean, variance, autocorrelation, cross-correlations
- **Uses in Quantitative Macro:**
 - *Calibration:* Match model moments to data moments
 - *Counterfactuals:* Study effects of policy changes
 - *Distribution computation:* Simulate agent distribution in HA models
- **Key Consideration:** Burn-in period; law of large numbers requires $T \rightarrow \infty$

Calibration: Mapping Models to Data

- **Goal:** Choose model parameters $\theta = (\beta, \alpha, \delta, \sigma, \rho, \dots)$ so the model matches key empirical regularities.
- **Calibration Strategies:**
 1. *Direct calibration:* Match parameter to single data moment directly
e.g., α = labor share ≈ 0.36 ; δ = depreciation rate ≈ 0.10
 2. *Indirect calibration:* Choose θ to minimize distance between model and data moments
e.g., β calibrated to match capital-output ratio
 3. *Steady-state calibration:* Use steady-state conditions as moment restrictions
- **Typical Moments Targeted:**
 - Capital-output ratio, consumption-output ratio
 - Labor share, average hours worked
 - Volatility and autocorrelation of GDP, consumption, investment
 - Cross-sectional wealth and income inequality

Estimation: Formal Inference

- **Calibration vs. Estimation:**

- Calibration: informal, one-to-one moment matching
- Estimation: formal statistical framework, uncertainty quantification

- **Simulated Method of Moments (SMM):**

$$\hat{\theta} = \arg \min_{\theta} [m_{\text{data}} - m_{\text{model}}(\theta)]' W [m_{\text{data}} - m_{\text{model}}(\theta)]$$

where W is a weighting matrix; accounts for estimation uncertainty.

- **Indirect Inference (Gourieroux et al., 1993):**

- Fit auxiliary model to both data and simulated data
- Match the *auxiliary model parameters*
- Useful when data has complex time-series structure

- **Bayesian Methods:**

- Prior $p(\theta)$ + likelihood $p(\text{data}|\theta) \Rightarrow$ posterior $p(\theta|\text{data})$
- MCMC sampling; popular for DSGE models

Accuracy Analysis: How Good Is the Solution?

- **Problem:** Numerical methods produce approximations. How do we measure quality?
- **Euler Equation Residuals (Judd, 1992):**

$$EEE(k) = \left| 1 - \frac{\beta u'(c_{t+1}) F_k(k_{t+1}, z_{t+1})}{u'(c_t)} \right|$$

- Measures how well the computed policy satisfies optimality conditions
 - Interpretation: fraction of consumption ("dollar unit") wasted by approximation error
 - Target: $|EEE| < 10^{-4}$ (less than 0.01% error)
- **Den Haan-Marcet Test:** Statistical test for whether forecast errors are unpredictable — checks if optimality conditions hold on average.
- **Policy Function Comparison:** Compare solution across methods (VFI, EGM, projection) to verify agreement.

Perturbation Methods: Linearization

- **Idea:** Approximate the equilibrium around the deterministic steady state.
- **First-Order Perturbation (Log-linearization):**

$$\hat{x}_{t+1} \approx A\hat{x}_t + B\hat{z}_t$$

where $\hat{x}_t = \log x_t - \log x^*$ are log-deviations from steady state.

- Fast and tractable; the workhorse of DSGE estimation (Dynare)
 - Ignores risk—agents behave as if the future is certain
- **Higher-Order Perturbation:**
 - 2nd order: captures precautionary saving, risk premia
 - 3rd order: skewness, asymmetric responses to shocks
 - Dramatically more complex, but still closed-form
- **Limitation:** Valid only near steady state; fails for large shocks or crises.

Supplementary Resource: Classical Numerical Methods

Video Lecture Series

Complete recordings of classical computational methods for macroeconomists are available at:

<https://space.bilibili.com/2142649036/lists/7180709>

Topics include:

- Numerical solution methods: value function iteration, time iteration
- Advanced methods: Howard improvement, endogenous grid method (EGM)
- Perturbation methods (first- and second-order)
- Projection methods: Chebyshev polynomials, collocation
- Numerical optimization, interpolation, and simulation
- High-performance and parallel computing (MPI, GPU)